Mehdi Karami

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Summary

– **Machine Learning Researcher** specializing in **image processing**, proposed a **pre-trained GAN for limited-label brain MRI segmentation** to advance **medical AI research**.

– Strong foundation in **machine learning theory and practice**, with experience in **model development, training, and optimization**, and a clear focus on contributing to **cutting-edge AI projects** and **real-world applications**.

– **Active open-source contributor** in AI, collaborating on **public and volunteer projects**, reflecting commitment to **innovation, collaboration, and advancement of the AI community**.

Education

2021–2025 **Bachelor of Computer Engineering**
Zand Institute of Higher Education
Shiraz, Persia
GPA: 17.88/20 (equivalent to 3.57/4.00)
Research: Medical Image Processing, Brain Image Segmentation

2023–2024 **Visiting Student (Bachelor of Computer Engineering)**
University of Science and Culture
Tehran, Persia
Bachelor's Project: AI Sight Quest (Integration of BERT-SQuAD for on-device document question answering)

Research Interests

– Machine learning
– Computer Vision
– Deep Learning
– Computational Biology
– Natural Language Processing
– Scientific Computing

Publication

Pre-Trained Generative Adversarial Network for Limited-Labeled Brain MRI Segmentation engrXiv, DOI: 10.31224/6504
Mehdi Karami✉, Betsabeh Tanoori

Open Source Projects

[1] **Retrieval-Augmented Generation (RAG) for Small Language Models (SLMs)**

– Implements a RAG pipeline which includes text chunking, embedding, and vector retrieval to enable efficient generation and inference with local Small Language Models (SLMs) as well as standard language models.

– Stack: Pytorch, FAISS, Transformers (HuggingFace), Sentence-Transformers, LangChain, PyMuPDF4LLM, ...

[2] **Cyberattack Detection and Anomalous Behavior Analysis**

– Utilizes an LSTM-based RNN to detect cyberattacks from anomalous network traffic, including DDoS, port scanning, and brute-force attempts.

– Stack: TensorFlow/Keras, Scikit-learn, PCA, ROC Curve, ...

[3] **Solar Power Generation Prediction**

– Implements ML models (Linear, Decision Tree, Random Forest, Gradient Boosting, MLP, DNN) to predict solar power generation, evaluated via MAE, MSE, R^2, and confusion matrix.

– Stack: TensorFlow, Scikit-learn, Pandas, NumPy, Seaborn, ...

[4] **NexumAI**

– Provides an interface for training models (e.g., image classification, segmentation) without technical expertise, letting users obtain trained models by supplying datasets and configurations.

– Stack: TensorFlow, PyTorch, Torchvision, PyQt6, ...

Additional projects can be found at: GitHub account .

Work Experience

2023–2024 **iOS Developer** (Full-time, on-site)
Round Table Apps (Sydney, Australia)
Development Office: Tehran, Persia
Working Projects:

- **Blossom – Save and Invest** , a financial application for the **Australian** and **New Zealand** markets, serving **25K+** active users.
- **MailPlus – MP Operator** , an **Australian** post-transportation application, serving **300+** owner-operators and **30K+** customers.
- **Resilience Box** , an **international** wellbeing and resilience platform.

Technical Contributions:

- Refactored legacy applications to improve maintainability, streamline **API integration**, and support future updates.
- Analyzed and implemented **data flows** to model **user behavioral patterns** across various scenarios.
- Applied key design patterns such as **MVC**, **MVVM**, **VIP**, and **Repository**.
- Research and development of **AR** and **VR** technologies in **healthcare** using **visionOS**.
- Boosted development efficiency by up to **40%** through reusable templates for modules and repository-pattern implementations.

Teaching Experience

Winter 2025 **Teaching Assistant (TA), Zand Institute of Higher Education**

- Fundamentals of Computer and Programming
- Digital Electronics
- Computer Architecture

Fall 2024 **Teaching Assistant (TA), Zand Institute of Higher Education**

- Human-Computer Interaction
- Digital Electronics

Technical Skills

- **Programming & Scripting:** Python, C/C++, Julia, MATLAB, Swift, Objective-C, SQL, Bash, LaTeX, ...
- **Data Science & ML:** TensorFlow/Keras, PyTorch, Scikit-learn, OpenCV, HuggingFace, NumPy, Pandas, ...
- **Deep Learning:** Neural Networks, Image Processing, Natural Language Processing, Transformers, GAN, RAG
- **Software Development:** Git, CI/CD, Design Patterns, SOLID, iOS Development, Unit Testing, TDD, BDD, Agile
- **Tools & Platforms:** Jupyter, Google Colab, GitHub, GitLab, Command Line, Linux, macOS, Jira, Confluence, Trello
- **Databases & Cloud:** MLOps, Vector DBs, Relational DBs (SQLite, PostgreSQL), Docker, Kubernetes, Query Optimization

Volunteering

Dec 2024 **Workshop on Git and GitHub, Zand Institute of Higher Education**, Certificate

May 2024 **Workshop on Git and GitHub, K. N. Toosi University of Technology**, Certificate

Honors & Awards

May 2022 **First Place Winner in the C++ Programming Language Competition, Zand Institute of Higher Education**

Hobbies

- Swimming
- Bodybuilding
- Traveling
- Gaming